CodyTolene / **pip-boy-3000-holotapes**

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Watch 0

A repository for Pip-Boy 3000 apps & games created by the community, and hosted on Pip-Boy.com

www.pip-boy.com

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Contributing

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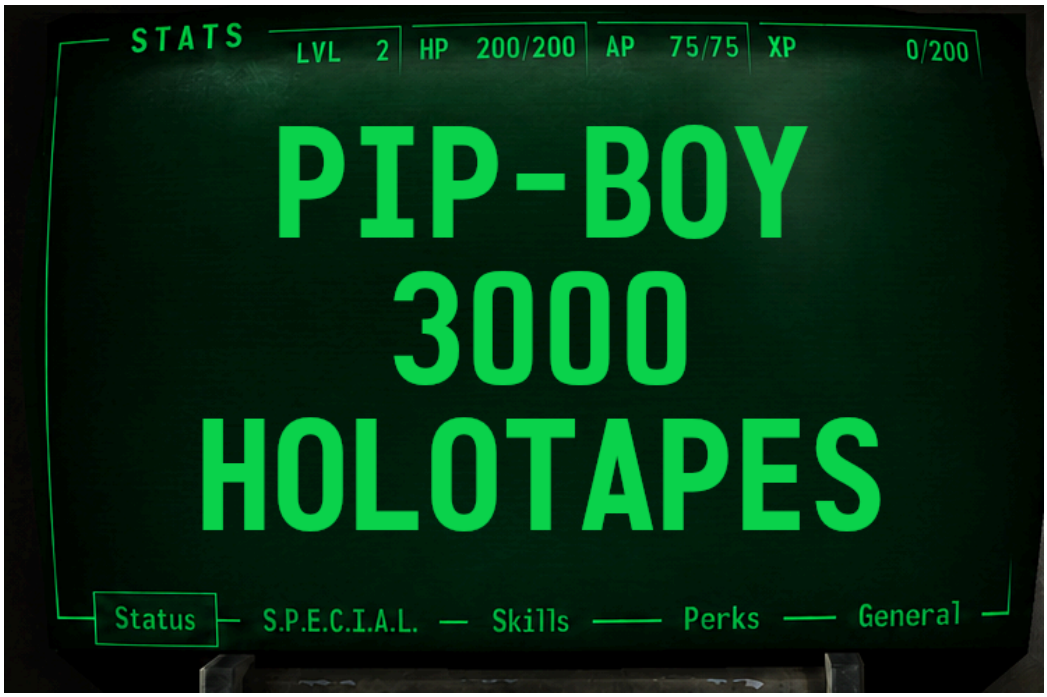
CodyTolene Reorganize preview images, menu first 5b1add6 · 17 hours ago

.github	Update refs	2 months ago
.husky	Add build scripts w/husky	2 months ago
.scripts	Fix up registry build	yesterday
.vscode	Add build scripts w/husky	2 months ago
holotapes	Reorganize preview images, menu f…	17 hours ago
.gitignore	Add build scripts w/husky	2 months ago
.prettierignore	Add build scripts w/husky	2 months ago
CODEOWNERS	iPip (v2.0.0) > RobCo Entertainment…	3 days ago
CONTRIBUTING.md	Documentation Updates (#47)	last week
LICENSE	Documentation Updates (#47)	last week
README.md	Documentation Updates Rev 2 (#49)	last week
agents.md	Documentation Updates Rev 2 (#49)	last week
package-lock.json	Update refs	2 months ago
package.json	Update refs	2 months ago
prettier.config.cjs	Add build scripts w/husky	2 months ago
	Add build scripts w/husky	2 months ago

tsconfig.json



Pip-Boy 3000 Holotapes

A community driven repository of custom applications and games for the Pip-Boy 3000, hosted on pip-boy.com.

Pip-Boy.com | Discord Community | Bethesda Store | The Wand Company | Espruino | RobCo Industries

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Description

Pip-Boy 3000 Holotapes by the community, for the community.

Install on: pip-boy.com

Follow the guide below to create your own custom Holotapes for the Pip-Boy 3000!

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Creating a new Holotape

1. Create a directory for the app or game under `holotapes/`. Every Holotape must use this structure:

```
holotapes/<YourHolotape>/
├── app.js
├── app.min.js
├── metadata.json
├── README.md
├── ChangeLog
└── assets/
```

2. Write the unminified source in `app.js`. The app must be an anonymous function expression that the Pip-Boy OS invokes; do not invoke it yourself with a trailing `()`. It must return an uppercase alphanumeric `id` and a `remove` function.

   Example app:

   ▶ Expand/Collapse

3. Add `metadata.json`. Asset paths and storage URLs are relative to the Holotape directory. The storage directory is the metadata `id` converted to uppercase, with underscores replacing hyphens.

```
{
  "id": "example",
  "name": "Example Holotape",
  "author": "@your-github-username",
  "version": "1.0.0",
  "description": "A short, one-sentence description.",
  "icon": "assets/icon.png",
  "previews": [],
  "type": "app",
  "readme": "README.md",
  "storage": [{ "name": "HOLO/EXAMPLE/APP.JS", "url": "app.min.js" }],
  "storageOptional": []
}
```

   The metadata `id` must contain only lowercase letters, numbers, and hyphens; `version` must use semantic versioning; and `type` must be exactly `app` or `game`. Do not edit `holotapes/registry.json` manually. `npm run build` generates it from each `metadata.json` file.

4. Document the description, controls, installation, tested firmware, and credits in the Holotape's `README.md`.

5. Add at least one `ChangeLog` entry in this format:

```
1.0.0 (yyyy-mm-dd)
<pull-request-or-change-link>
- Initial release
```

6. Generate `app.min.js` from `app.js` as described in Build and minification. Both files must behave identically.

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Development Workflow

Using Pip-Boy.com

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Using the Espruino Web IDE

You can use one of the two methods below to upload and test your Holotape:

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Build and minification

Minify and Encode your Holotape here:

https://www.pip-boy.com/3000/holotapes/create

This will give you a proper `app.min.js` from `app.js` .

Run the repository build after adding or changing metadata:

```
npm install
npm run build
```

This rebuilds `holotapes/registry.json` , rewrites relative file paths for the registry, and rejects metadata whose `type` is not `app` or `game` . It does not generate `app.min.js` or perform all of the submission checks for you.

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Images

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Input handling

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Memory and Performance

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Main rules:

- The display is always 480×320. If dimensions are needed repeatedly, declare them once as `const W =`

never `var`, and inline single-use values.

- Always use the global `h` graphics object directly and chain graphics calls. Do not spend a variable block on an alias for `h`.
- Use `h.clearRect()`, clipping, cached wrapped text, and dirty flags to redraw only changed regions.
- Timer-driven apps normally rely on the OS auto-flush. Do not call `h.flip()` from a `setInterval()` loop. For `Pip.onFrame` rendering, set `Pip.lastFlip = getTime()` before drawing and call `h.flip()` after drawing.
- Use `"ram"` for performance-critical frame functions and `"jit"` for small, numeric tight loops. Do not use either without a measured need.
- Use `Math.randInt(n)` instead of `Math.random()`, and use typed arrays for dense numeric data.
- Espruino does not support `async` / `await`, ES modules, template literals, `fetch()`, `XMLHttpRequest`, or `requestAnimationFrame()`.
- Keep the app scoped:

```
(function () {
  // App code here
  return {
    id: 'APPID',
    notDefault: true,
    fullscreen: true,
    remove: function () {
      /* clean up app resources */
    },
  };
});
```

- Clean up every resource created by the app in `remove()`:

```
  remove: function() {
    Pip.removeListener("knob1", onKnob1);
    clearInterval(intervalId);
    clearTimeout(timeoutId);
    clearWatch(watchId);
    Pip.audioStop();
    h.clear();
  }
```

- Never call `load()` or `E.reboot()` from `remove()`, call `Pip.remove()` at the top of the app, use a bare `clearWatch()`, or delete/reassign OS globals.

Useful memory checks:

```
process.memory();
print(E.getSizeOf(this, 1).sort((a, b) => a.size - b.size));
print(E.getSizeOf(Pip, 1).sort((a, b) => a.size - b.size));
print(E.getSizeOf(this['\xFF'], 1).sort((a, b) => a.size - b.size));
```

`this['\xFF']` shows timers, watches, and internal runtime state.

`Pip.CURRENT` can hold the current page or app code.

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Contributing

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By submitting a pull request or contribution, you agree that your contribution is provided under the MIT License. See CONTRIBUTING.md for details.

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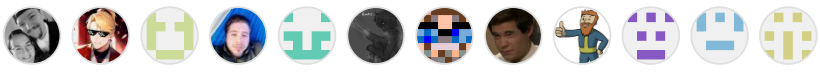

Languages